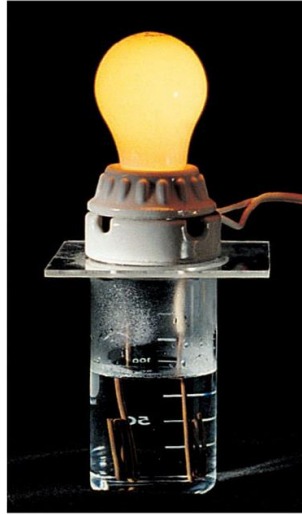


CH 11- Solutions & Colloids –Pt3

- Introduction
- 11.1 The Dissolution Process
- 11.2 Electrolytes
- 11.3 Solubility
- 11.4 Colligative Properties
- ~~11.5 Colloids~~



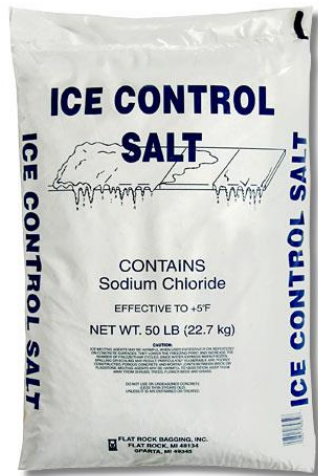
strong electrolyte



weak electrolyte



nonelectrolyte



Colligative Properties of Solutions

- **Colligative properties** are properties that depend on the **number** of solute particles, not their chemical identity.
- The number of particles in solution can be predicted from the formula and type of the solute.
- An **electrolyte** separates into ions when it dissolves in water.
 - Strong** electrolytes dissociate completely while **weak** electrolytes dissociate very little.
 - i takes on values other than 1***
- A **nonelectrolyte** does not dissociate to form ions.
 - Each mole of dissolved compound yields 1 mole of particles in solution. ***$i = 1$***

For vapour pressure lowering: $\Delta P = i(X_{\text{solute}} \times P^{\circ}_{\text{solvent}})$

For boiling point elevation: $\Delta_b T = i(K_b m)$

For freezing point depression: $\Delta_f T = i(K_f m)$

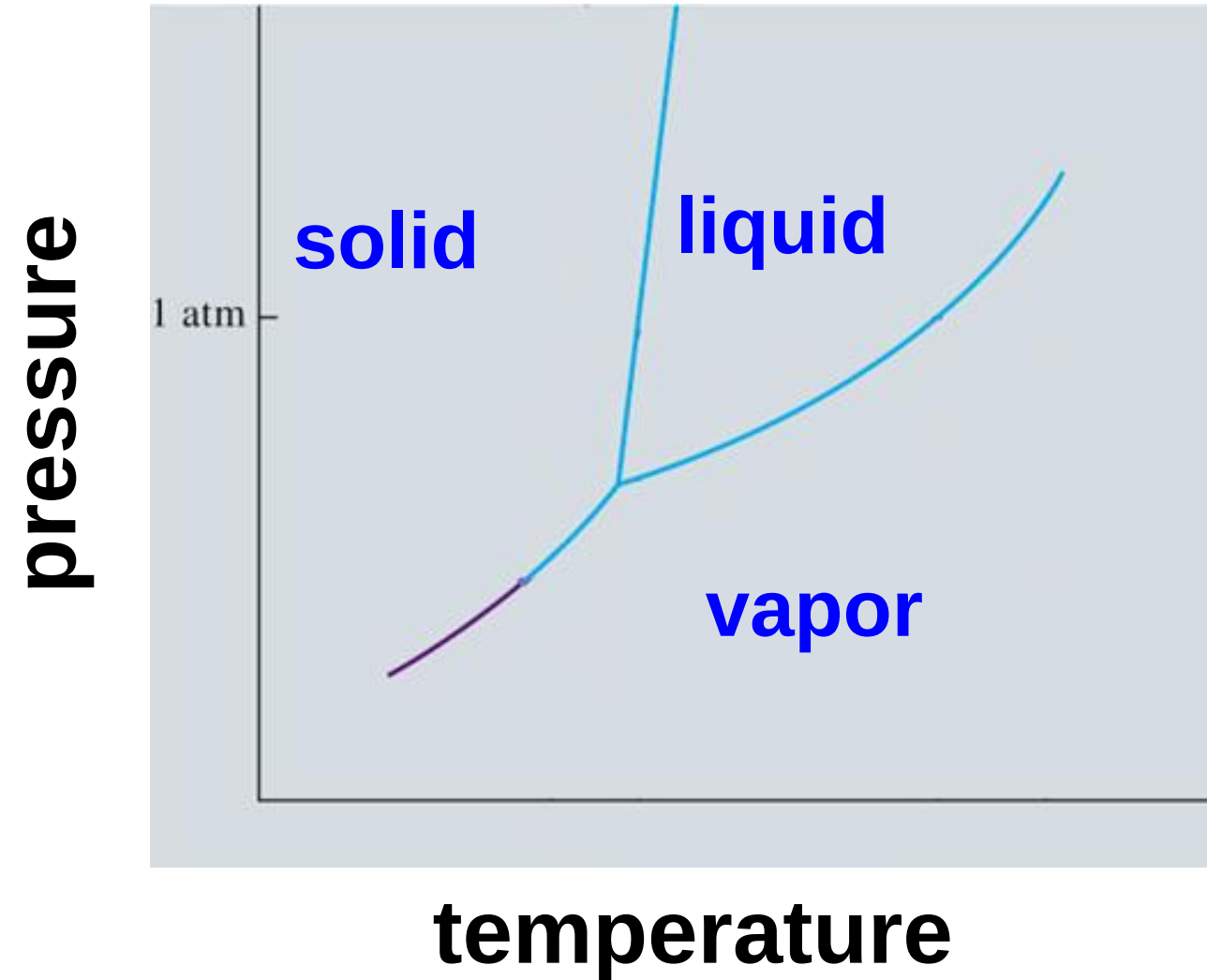
For osmotic pressure: $\Pi = i(cRT)$

Vapour Pressure Lowering, Freezing Point Depression and Boiling Point Elevation

- Dissolved nonvolatile solutes:
 - lower the freezing point of the solvent.
 - If enough solute is added, freezing point of water can be lowered below the ambient temperature: ice melts.
 - lower the vapor pressure of the solvent
 - increase the boiling point of the solution.

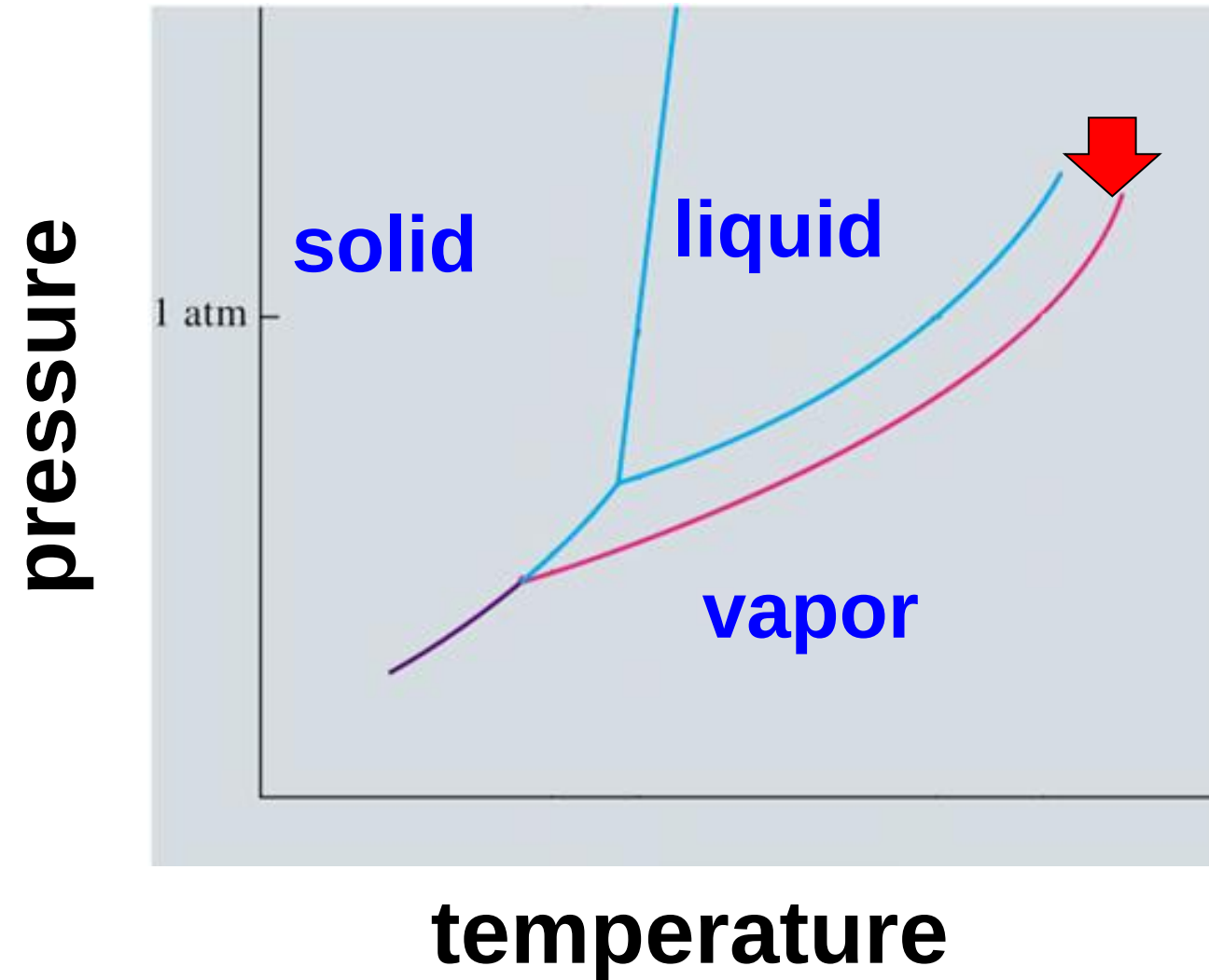
$$P_A = x_A P_A^\circ$$

Phase Diagram



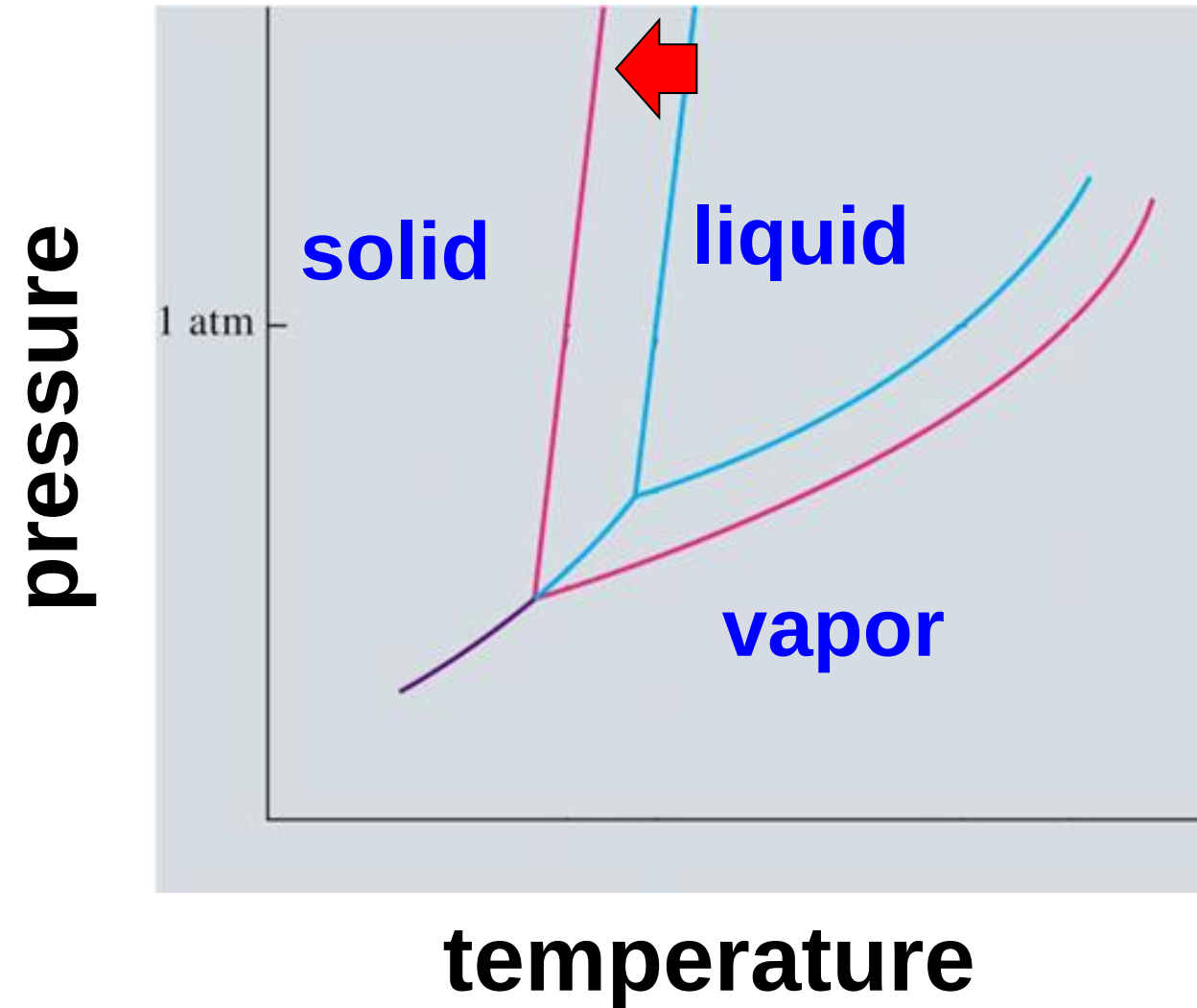
- Recall the typical phase diagram

Phase Diagram, cont'd



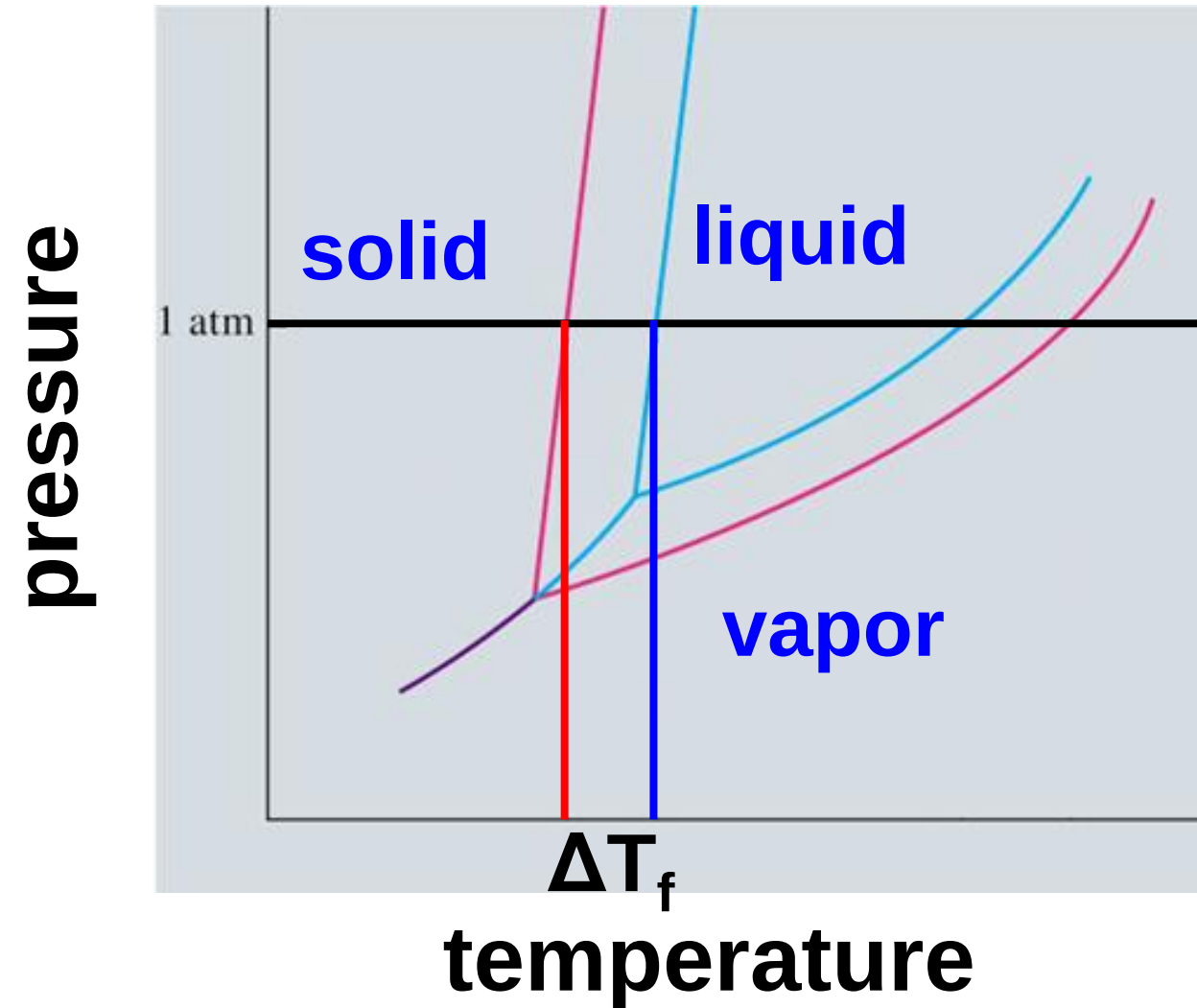
- Addition of a solute reduces the vapor pressure.
- Shifts the vapor curve to lower pressures (down to the right).

Phase Diagram, cont'd



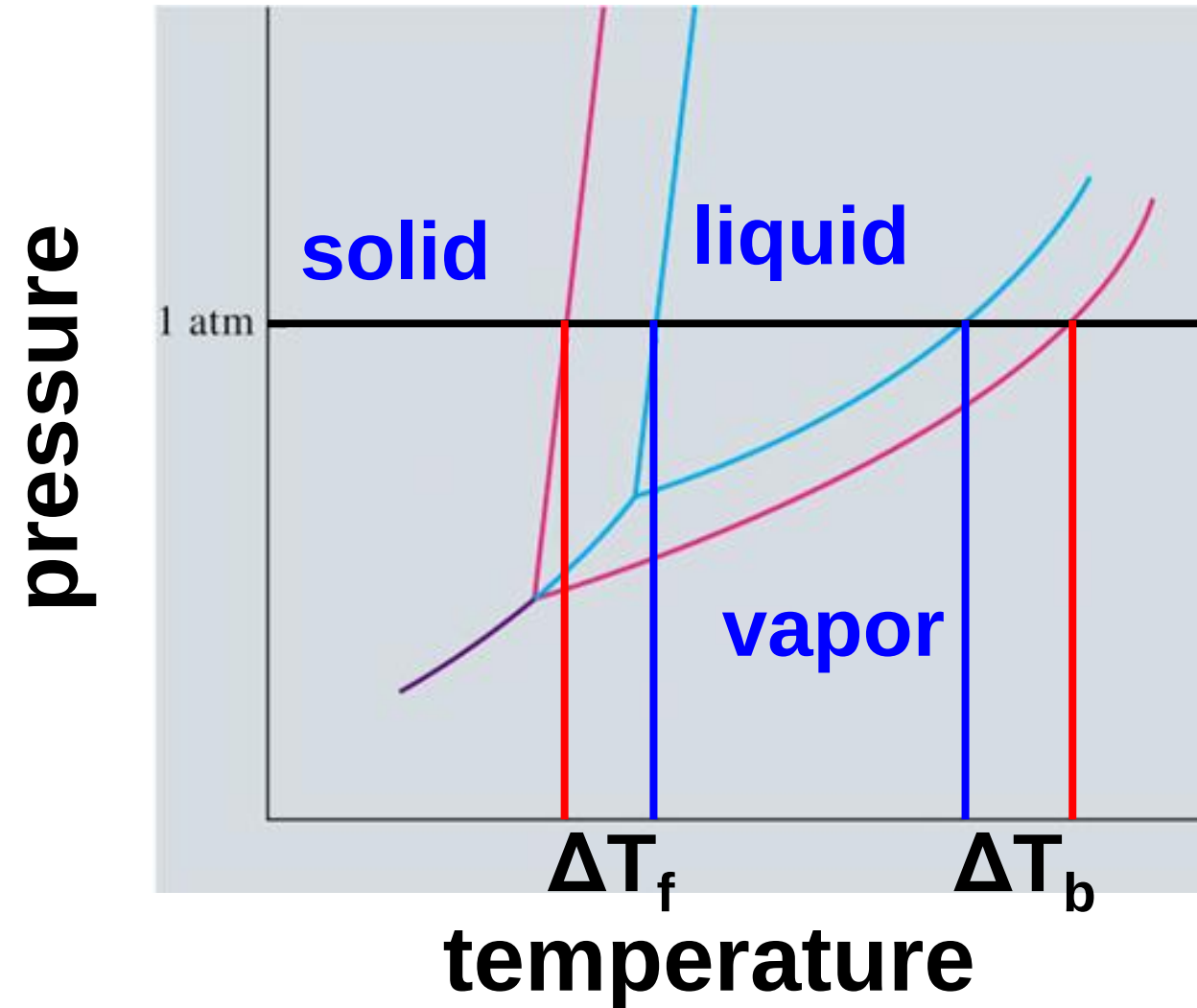
- Addition of a solute shifts the fusion curve to lower temperatures
- (to the left)

Phase Diagram, cont'd



- Intersection of fusion curve with 1.0 atm gives the **normal melting point**.
- Shifts down by ΔT_f

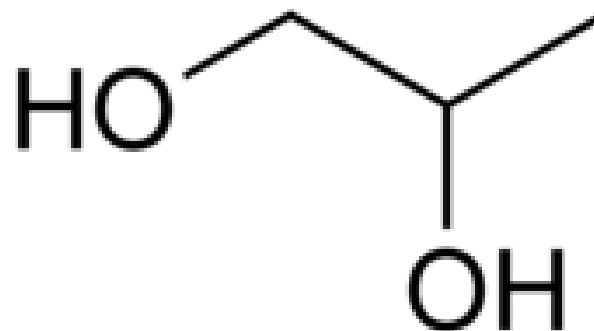
Phase Diagram, cont'd



- Intersection of vapor curve with 1.0 atm gives the **normal boiling point**.
- Shifts up by ΔT_b

Anti-freeze

Propylene glycol



- Prevents freezing in winter.
- Prevents boiling in summer.



Non-Electrolyte Solutions

$$\begin{aligned}\Delta T_f &= -K_f * m \\ \Delta T_b &= +K_b * m\end{aligned}$$

Concentration of solute (molality)

- Shifts in freezing and boiling points depend linearly on the concentration of solute.
- K_f and K_b vary with the solvent.
- K_f and K_b do not depend on solute (non-electrolyte).
- Equations valid for dilute solutions.

Molecular Weights

$$\Delta T_f = -K_f * m$$

$$\Delta T_b = +K_b * m$$

- Dissolve mystery solute in a known solvent.
- Measure ΔT_f or ΔT_b .
- K_f and K_b are known.
- Determine concentration, m .
- Mass of dissolved is known.
- Calculate molecular weight.

Molal Boiling Point Elevation and Freezing Point Depression Constants of Several Solvents

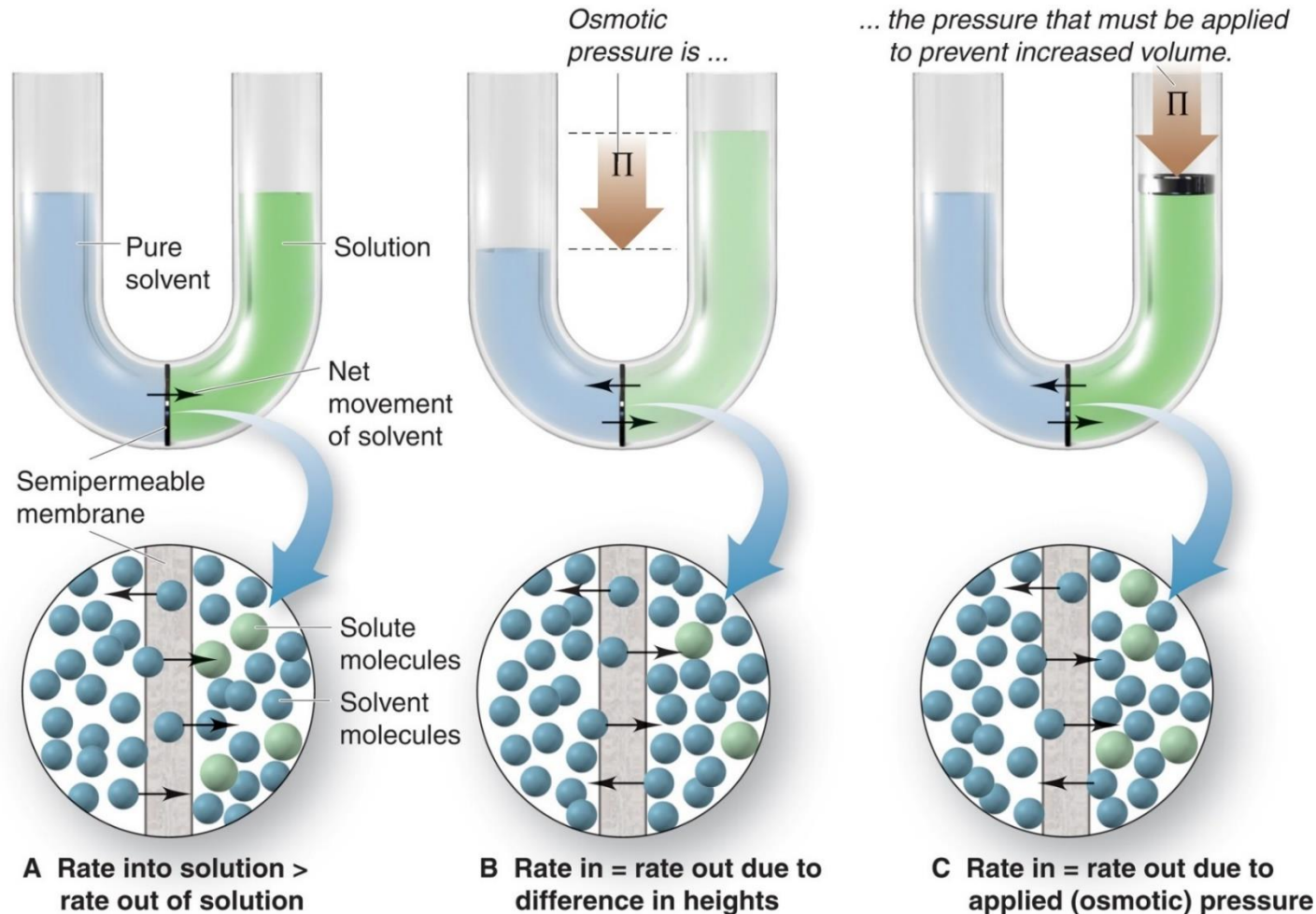
Boiling Point Elevation and Freezing Point Depression Constants for Several Solvents

Solvent	Boiling Point (°C at 1 atm)	K_b (°C m^{-1})	Freezing Point (°C at 1 atm)	K_f (°C m^{-1})
water	100.0	0.512	0.0	1.86
hydrogen acetate	118.1	3.07	16.6	3.9
benzene	80.1	2.53	5.5	5.12
chloroform	61.26	3.63	-63.5	4.68
nitrobenzene	210.9	5.24	5.67	8.1

Table 11.2

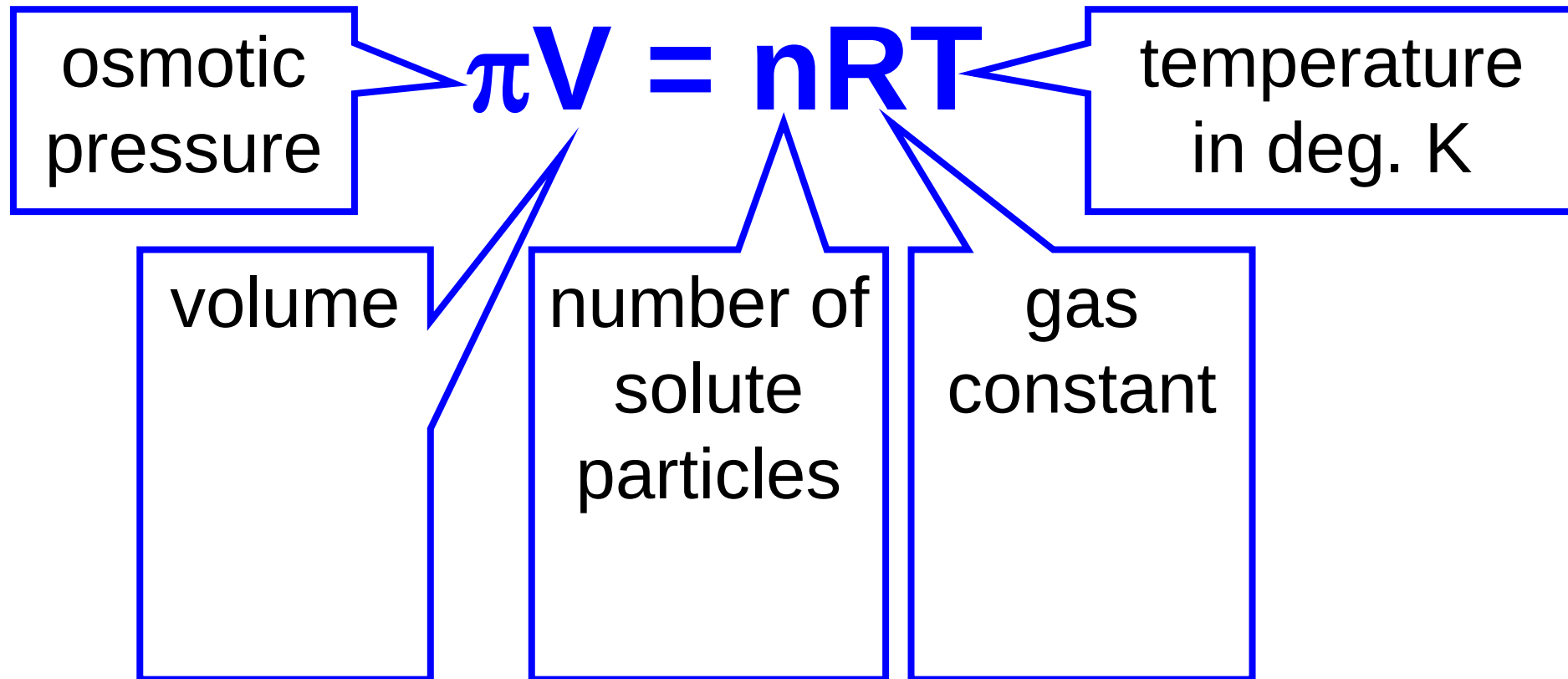
Osmotic Pressure

- Diffusion across semi-permeable membranes.



Osmotic Pressure, cont'd

- Osmotic pressure depends on the concentration of solute particles.
 - similar to ideal gas equation



Osmotic Pressure, cont'd

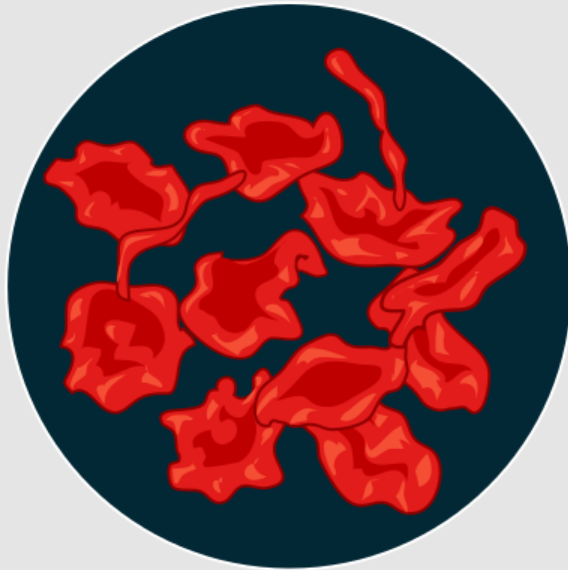
$$\pi = \frac{n}{V} RT$$

concentration, **molarity**

$$\pi = M*RT$$

Osmotic Pressure and Cell Shape

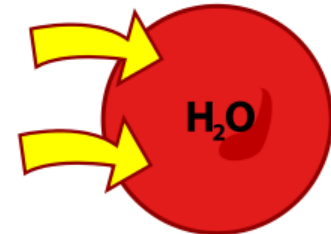
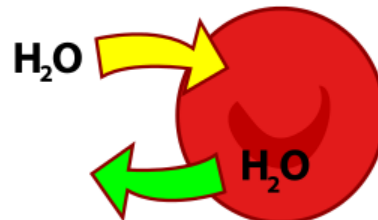
Hypertonic



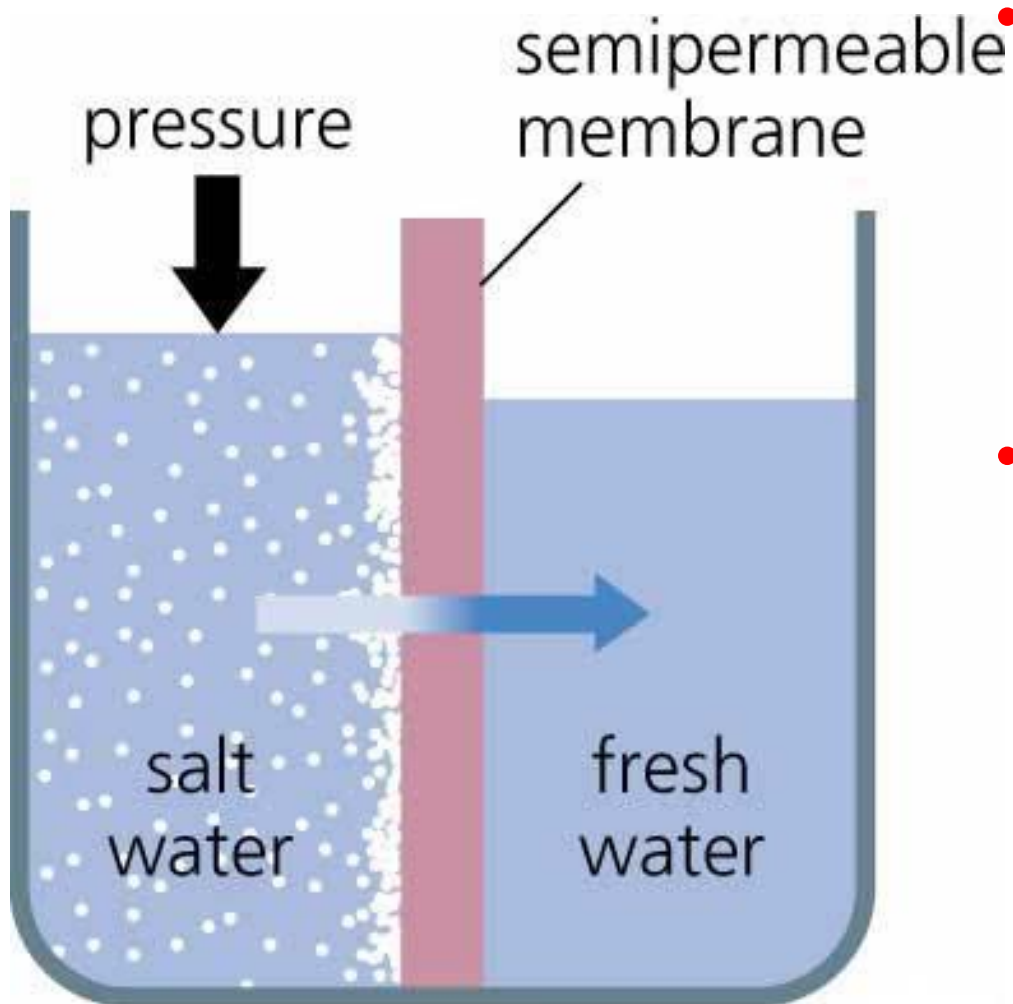
Isotonic



Hypotonic

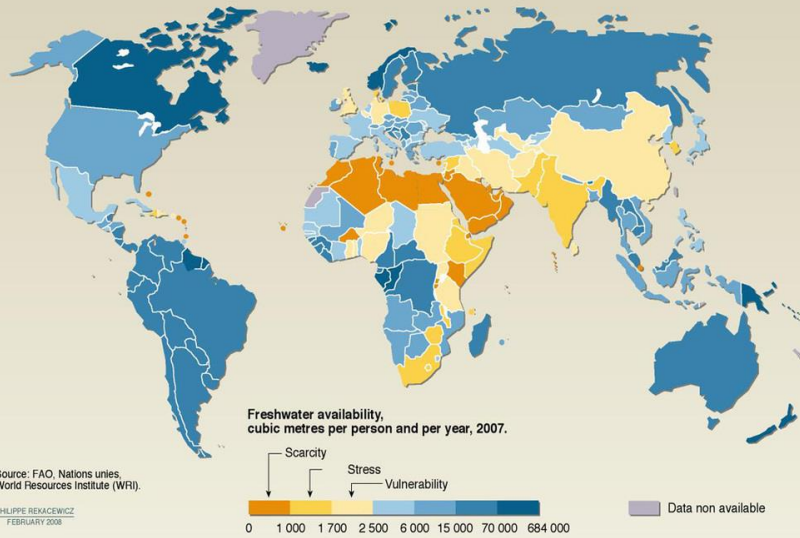


Reverse Osmosis

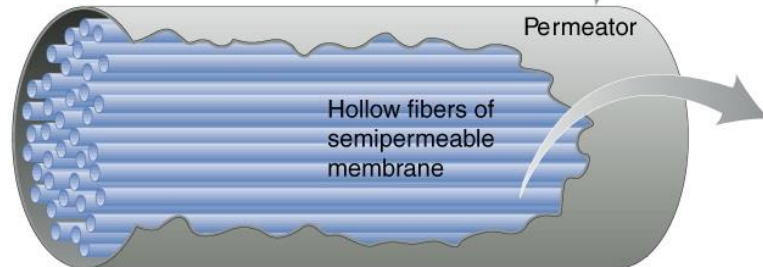


- Suppose you apply a pressure **greater than the osmotic pressure**.
- You force H_2O from the solute side to the pure water side, i.e., you can remove H_2O from the solute/ H_2O solution.

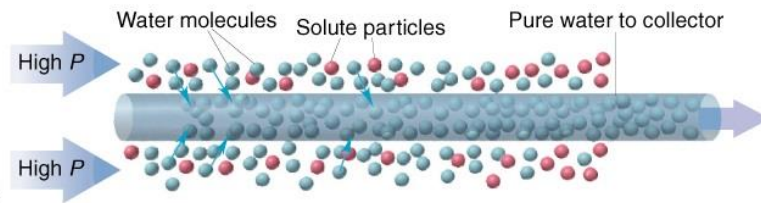
Desalination is a potential source of freshwater from Ocean Water.



A



B



C

Colligative Properties of Electrolyte Solutions

Strong Electrolyte Solutions

- A strong electrolyte dissociates completely to form ions. Each mole of solute gives more than 1 mol of dissolved particles.
- The *formula* of the compound indicates the expected number of particles in solution.
 - Each mol of NaCl is expected to give 2 moles of dissolved ions,
- The *van't Hoff factor* takes into account the dissociation of a strong electrolyte to predict the effect on the solution.

$$i = \frac{\text{measured value for electrolyte solution}}{\text{expected value for nonelectrolyte solution}}$$

Electrolyte Solutions

- By how much would a 0.0100 molal solution of a non-volatile substance be expected to lower the freezing point of water?

$$\Delta T_f = - K_f * m$$

$$= -1.86 \text{ } ^\circ \text{ C} \cdot \text{kg/mol} * 0.0100 \text{ mol/kg}$$

$$= -0.0186 \text{ } ^\circ \text{ C}$$

Electrolyte Solutions, cont'd

For NaCl:

- Expected depression: $-0.0186\text{ }^{\circ}\text{C}$
- Measured depression: $-0.0361\text{ }^{\circ}\text{C}$

We can define the **van't Hoff factor**:

$$i = \frac{\text{measured } \Delta T_f}{\text{expected } \Delta T_f} = \frac{-0.0361^{\circ}\text{C}}{-0.0186^{\circ}\text{C}} = 1.94$$



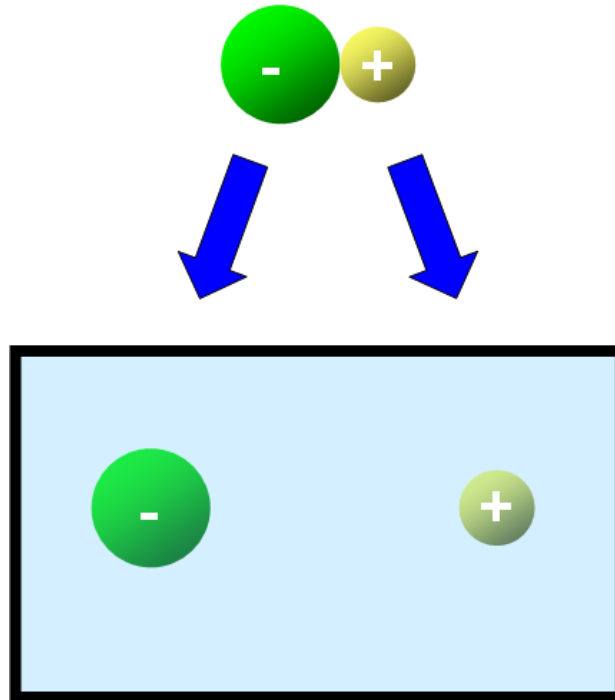
Jacobus Hendricus
van 't Hoff

(1852-1911)

J. H. van 't Hoff.

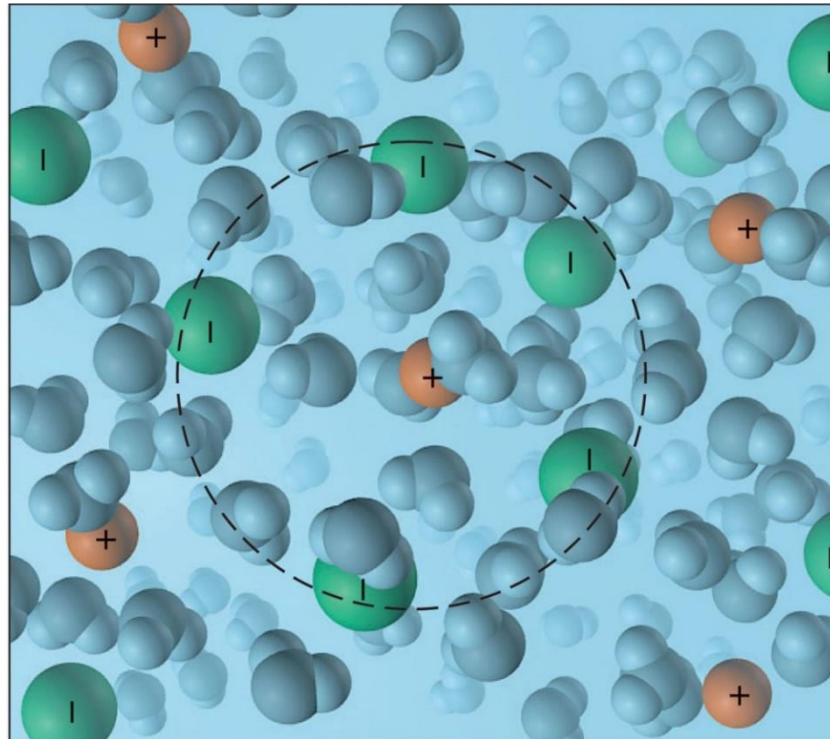
Electrolyte Solutions, cont'd

- Colligative effects of NaCl are close to double: $i = 1.94$
 - 2 moles of dissolved particles per mole of NaCl.



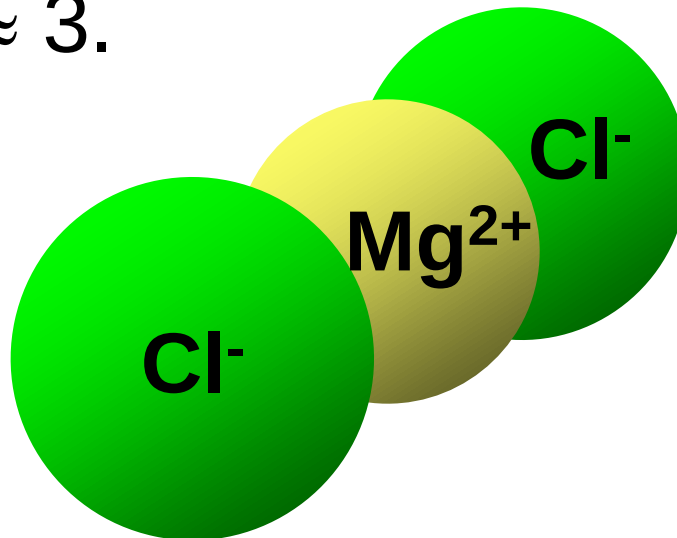
Electrolyte Solutions, cont'd

- Why is $i < 2$?
- Some weak interactions remain between the hydrated ions.
 - i is lower for concentrated solutions.
 - i approaches 2 for infinitely dilute solutions.



Electrolyte Solutions, cont'd

- i can be thought of as the effective number of moles of independent dissolved particles produced by one mole of solute.
- For non-electrolytes, $i = 1$.
- For MgCl_2 , $i \approx 3$.



Colligative Properties

- We can recast the colligative equations for electrolytes:

$$\pi = i * M * R T$$

$$\Delta T_f = -i * K_f * m$$

$$\Delta T_b = i * K_b * m$$

Predicated and Measured van't Hoff Factors for Several 0.050 *m* Aqueous Solutions

Formula unit	Classification	Dissolution products	<i>i</i> (predicted)	<i>i</i> (measured)
C ₁₂ H ₂₂ O ₁₁ (glucose)	Nonelectrolyte	C ₁₂ H ₂₂ O ₁₁	1	1.0
NaCl	Strong electrolyte	Na ⁺ , Cl ⁻	2	1.9
HCl	Strong electrolyte (acid)	H ₃ O ⁺ , Cl ⁻	2	1.9
MgSO ₄	Strong electrolyte	Mg ²⁺ , SO ₄ ²⁻ ,	2	1.3
MgCl ₂	Strong electrolyte	Mg ²⁺ , 2Cl ⁻	3	2.7
FeCl ₃	Strong electrolyte	Fe ³⁺ , 3Cl ⁻	4	3.4

Table 11.3

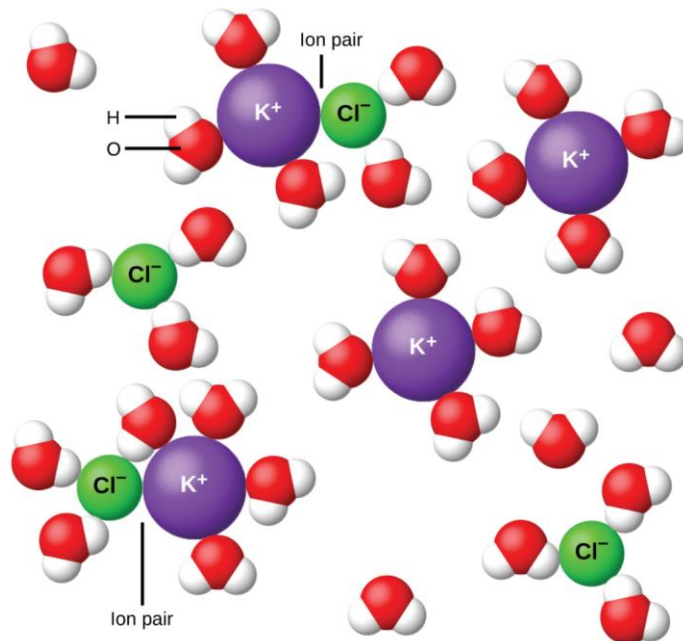


Figure 11.28 Dissociation of ionic compounds in water is not always complete due to the formation of ion pairs.